

Unit 6.2 Vectors

Note Title

2015/02/19

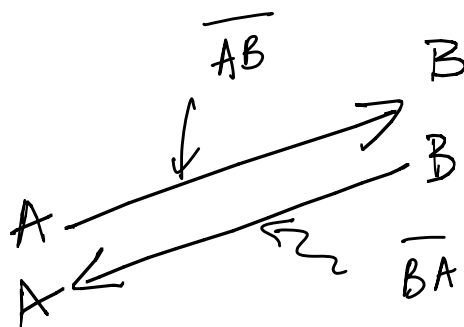
A vector

- has magnitude and direction
- does not have a fixed position

Notation

- A - point, B - point

$\overrightarrow{AB}$ vector, $\overrightarrow{BA}$ vector



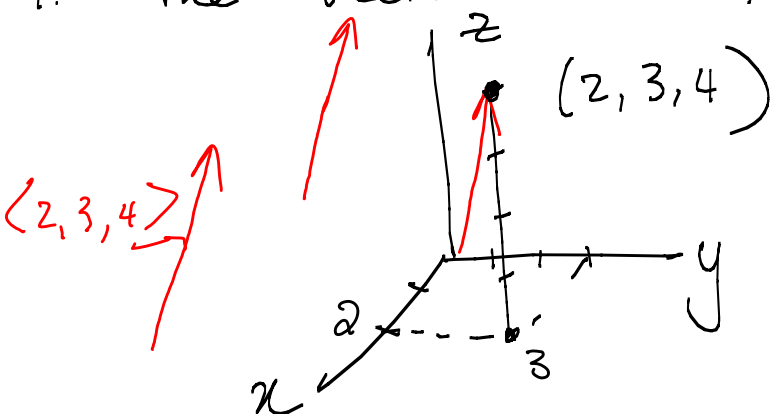
$$\vec{a} = \langle a_1, a_2, a_3 \rangle \text{ in } \mathbb{R}^3$$

$$\vec{a} = \langle a_1, a_2 \rangle \text{ in } \mathbb{R}^2$$

() point

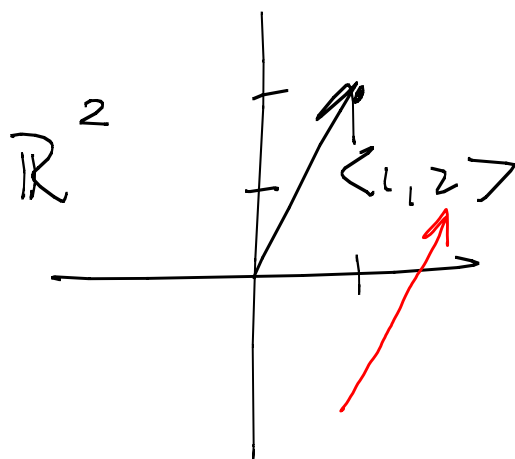
Examples

1. The vector $\langle 2, 3, 4 \rangle$ in $\mathbb{R}^3$



Departs from origin - position vector

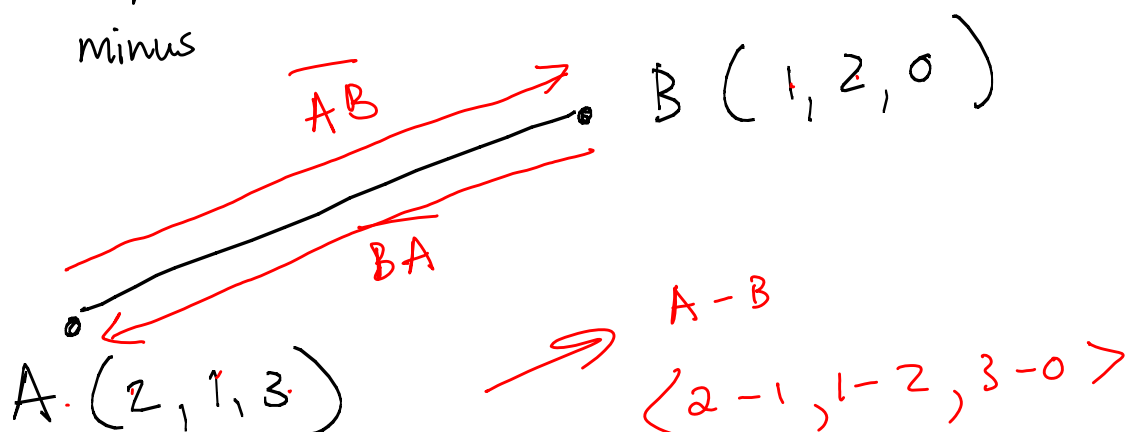
2. The vector $\langle 1, 2 \rangle$ in $\mathbb{R}^2$



Vector between two points

endpoint $\xrightarrow{\quad}$ starting point

minus



$$\overrightarrow{BA} = \langle 1, -1, 3 \rangle$$

st. $\swarrow$ $\searrow$ end.

$$\overrightarrow{AB} = -\overrightarrow{BA} = \langle -1, 1, -3 \rangle.$$

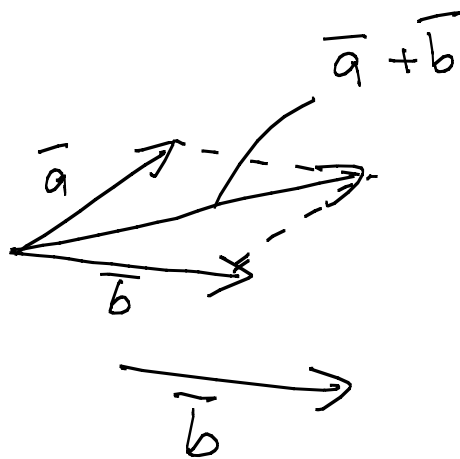
operations with vectors:

Addition

$$\vec{a} = \langle a_1, a_2, a_3 \rangle$$

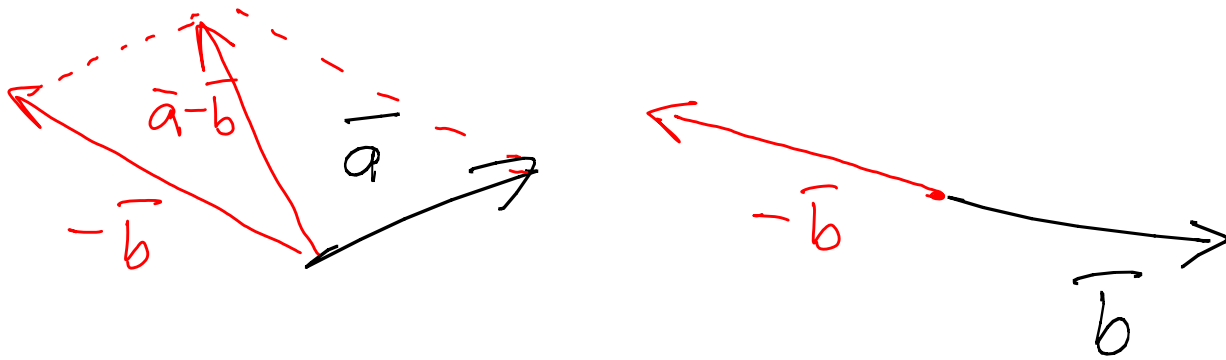
$$\vec{b} = \langle b_1, b_2, b_3 \rangle$$

$$\vec{a} + \vec{b} = \langle a_1 + b_1, a_2 + b_2, a_3 + b_3 \rangle$$



eg $\langle 1, 0, 2 \rangle + \langle 2, -1, 3 \rangle = \langle 3, -1, 5 \rangle$

Subtraction : $\vec{a} - \vec{b} = \vec{a} + (-\vec{b})$



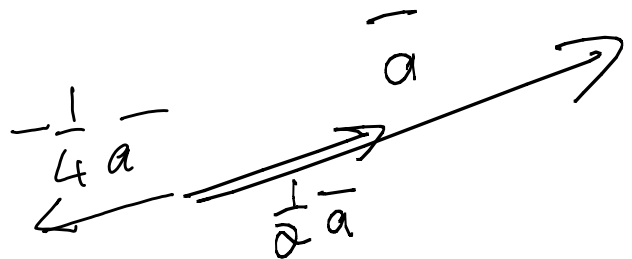
$$\langle 2, 3, 1 \rangle - \langle 1, 0, 2 \rangle = \langle 1, 3, -1 \rangle.$$

Scalar multiplication

$$\begin{aligned} r\vec{a} &= r \langle a_1, a_2, a_3 \rangle \\ &= \langle ra_1, ra_2, ra_3 \rangle \end{aligned}$$

Eg. $\frac{1}{2} \langle 2, 1, 0 \rangle = \langle 1, \frac{1}{2}, 0 \rangle$

$$-\frac{1}{4}\vec{a}$$



Parallel vectors

- Vectors $\vec{a}$ and $\vec{b}$ are parallel if there exists a number $c \in \mathbb{R}$ such that

$$\vec{a} = c\vec{b}$$

Examples

1. $\vec{a} = \langle 1, 2, -2 \rangle \parallel \vec{b} = \langle 4, 8, -8 \rangle$ 

2. $\vec{a} = \langle 1, 2, -2 \rangle \parallel \vec{b} = \langle -\frac{1}{2}, -1, 1 \rangle$

1. $\vec{a} = \frac{1}{4}\vec{b}$

($\vec{b} = 4\vec{a}$)

2. $\vec{a} = -2\vec{b}$

($\vec{b} = -\frac{1}{2}\vec{a}$)

Length of a vector

$$|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

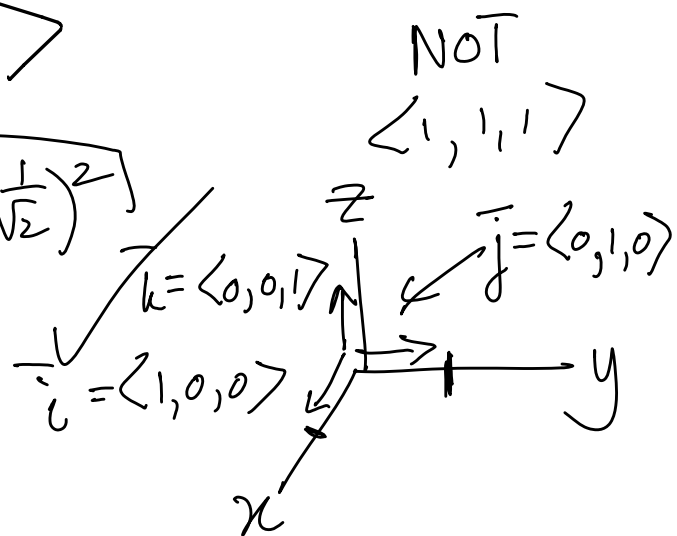
Eg. $|\langle 2, -1, 4 \rangle| = \sqrt{2^2 + (-1)^2 + 4^2} = \sqrt{21}.$

A unit vector

$\bar{a}$ is a unit vector if $|\bar{a}| = 1$.

eg. $\bar{a} = \langle \frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}} \rangle$

$$|\bar{a}| = \sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + 0^2 + \left(-\frac{1}{\sqrt{2}}\right)^2}$$
$$= 1.$$



A unit vector in the direction of $\bar{a}$.

$$\bar{b} = \frac{\bar{a}}{|\bar{a}|}$$

Diagram showing a vector $\bar{a}$ and its unit vector $\bar{b}$. The unit vector $\bar{b}$ is shown as a smaller vector in the same direction as $\bar{a}$, with a bracket indicating its length is 1.

WR $\frac{\bar{a}}{|\bar{a}|}$ is a unit vector in the direction of $\bar{a}$.

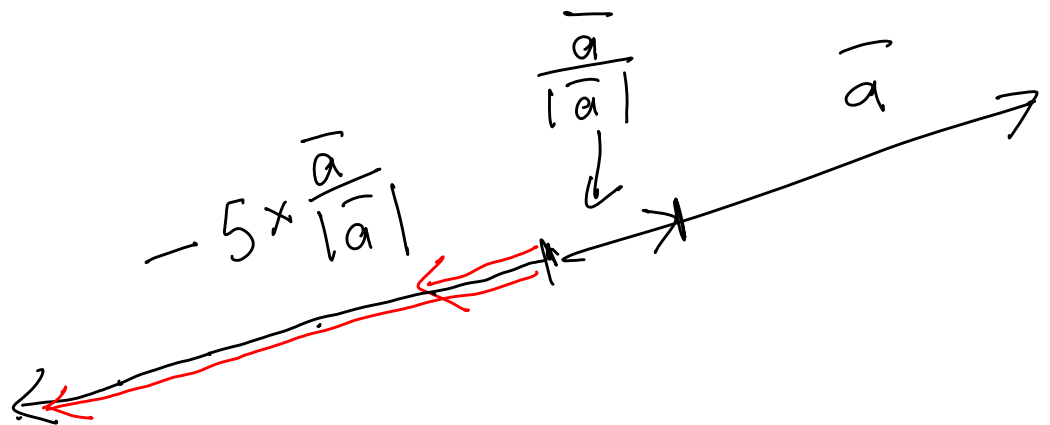
eg. Find the unit vector in the direction of

$$\bar{a} = \langle 2, -1, 3 \rangle$$

$$\frac{\bar{a}}{|\bar{a}|} = \frac{1}{\sqrt{14}} \langle 2, -1, 3 \rangle = \left\langle \frac{2}{\sqrt{14}}, -\frac{1}{\sqrt{14}}, \frac{3}{\sqrt{14}} \right\rangle$$

$$|\vec{a}| = \sqrt{2^2 + (-1)^2 + 3^2} = \sqrt{14}.$$

Ex. Find a vector of length 5 in the opposite direction to $\vec{a} = \langle 1, -2, 1 \rangle$



$$|\vec{a}| = \sqrt{1+4+1} = \sqrt{6}.$$

$$\Rightarrow -5 \frac{\vec{a}}{|\vec{a}|} = \frac{-5}{\sqrt{6}} \langle 1, -2, 1 \rangle.$$